

```

1  from collections import deque
2
3  def water_jug():
4      visited = set()
5      queue = deque()
6
7      # Start with both jugs empty
8      queue.append((0, 0, []))
9
10     while queue:
11         a, b, path = queue.popleft()
12
13         if (a, b) in visited:
14             continue
15
16         visited.add((a, b))
17         path = path + [(a, b)]
18
19         # Goal condition: exactly (2,0)
20         if (a, b) == (2, 0):
21             return path
22
23         # Carefully ordered moves to
24         # match desired output
25         next_states = [
26             (4, b), # Fill 4L jug
27             (a, 3), # Fill 3L jug
28             (0, b), # Empty 4L jug
29             (a, 0), # Empty 3L jug
30             (a - min(a, 3 - b), b + min

```

Run

```

18
19     # Goal condition: exactly (2,0)
20     if (a, b) == (2, 0):
21         return path
22
23     # Carefully ordered moves to
        match desired output
24     next_states = [
25         (4, b), # Fill 4L jug
26         (a, 3), # Fill 3L jug
27         (0, b), # Empty 4L jug
28         (a, 0), # Empty 3L jug
29         (a - min(a, 3 - b), b + min
            (a, 3 - b)), # Pour A
                -> B
30         (a + min(b, 4 - a), b - min
            (b, 4 - a)), # Pour B
                -> A
31     ]
32
33     for state in next_states:
34         if state not in visited:
35             queue.append((state[0],
                state[1], path))
36
37     result = water_jug()
38
39     print("Steps:")
40     for step in result:

```

Run

Steps:

(0, 0)

(0, 3)

(3, 0)

(3, 3)

(4, 2)

(0, 2)

(2, 0)

=== Code Execution Successful ===